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ASSIGNMENT

Assignment

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1 Introduction

Over the past decade, the use of E-cigarettes has increased significantly. This rapid growth in usage is specially seen among young adults. At the same time, prediabetes has become a serious public health concern. Prediabetes is a condition where the patient's blood sugar level is higher than normal, but it is not high enough to be diagnosed as diabetes. Hence, it is very important to understand whether e-cigarette use is linked to prediabetes for preventing and improving public health outcomes.

2 Theory and Background

The condition where the blood sugar levels are higher than normal but not high enough to be diagnosed as diabetes is called Prediabetes. Generally, it is a warning, as it increases the risk of developing type 2 diabetes and other health problems. The prevalence of prediabetes has increased significantly in recent years, making it a major public health concern (Zhang et al., 2022).

The use of e-cigarettes has also risen rapidly among the youth. They are often promoted as safer alternative to traditional tobacco cigarettes but their long-terms health consequences are not fully comprehended (Zhang et al., 2022). Researchers have started to explore whether e-cigarette use may also affect metabolic health.

The primary explanation for this phenomenon is the presence of nicotine in e-cigarettes. The control of blood sugar levels by the body becomes disrupted because nicotine causes glucose levels to rise, while disrupting typical metabolic functions. Zhang et al. 2022 discovered a connection between the condition which causes body inflammation and various bodily changes that create a higher chance of developing prediabetes.

The research from earlier studies shows that traditional cigarette smoking leads to insulin resistance which increases the chance of developing diabetes (Willi et al., 2007). E-cigarette use shows a possible link to prediabetes according to findings from recent studies (Atuegwu et al., 2019). The previous research studies had limitations because they used smaller datasets and focused on particular demographic groups.

Due to these limitations, it becomes important to perform a comprehensive study on the relationship between e-cigarettes and prediabetes.

3 Aim of the study

The aim of the study was to understand the association between e-cigarette use and prediabetes among the adults of United States. The researchers used different groups of users: current users, former users, those who had never used e-cigarettes. They also aimed to determine whether this association remained consistent among sole e-cigarette users and dual users (Zhang et al., 2022).

4 Methodology

The data used in this study are from the Behavioural Risk Factor Surveillance System (BRFSS). It is a large national health survey conducted in the USA. The time range is 2016 to 2018. The final sample had 600,046 adults with age 18 years and above. Individuals with diabetes and missing data were excluded in the study.

The researchers used self-reported data to create three e-cigarette use categories which included present users, previous users and never users. The researchers divided participants into two group that included sole users and dual users based on their use of traditional cigarettes. The researchers classified participants as prediabetic when they reported receiving a medical diagnosis from a healthcare professional.

The researchers considered multiple additional factors which included age, gender, education level, body weight, physical activity level and existing health conditions. The researchers used logistic regression models to examine the relationship between e-cigarette use and prediabetes while controlling for these variables (Zhang et al., 2022).

5 Findings

The study showed a clear relation between the use of e-cigarettes and likelihood of prediabetes. Individuals who were currently using e-cigarettes were much more likely to report prediabetes in comparison to those who had never used them. Even former users show an increased likelihood of prediabetes.

Solo e-cigarette users showed the highest increase in risk when compared to individuals who had never used any tobacco products. Dual users also showed an increase in likelihood but the association was weaker than solo users.

In terms of overall patterns, the prevalence of prediabetes differed across groups. It was observed to be 9.0% among current e-cigarette users, 10.2% among dual users, and 5.9% among sole users. These differences suggest that e-cigarette use, even without traditional cigarette use, is linked to changes in metabolic health.

Overall, the results indicate that e-cigarette use is associated with an increased probability of developing prediabetes, independent of many other influencing factors (Zhang et al., 2022).

6 Connection to another research

The findings of this study are consistent with the previous studies and research in this field. Earlier studies established that traditional cigarette smoking can cause insulin resistance which brings a higher risk of developing diabetes (Willi et al., 2007). This study extends that research by showing that a similar risk can also be present in e-cigarette use.

The research results support existing research which showed that prediabetes connects to e-cigarette use in specific groups of people (Atuegwu et al. 2019). The study included only participants who had never smoked traditional cigarettes and used a smaller database to conduct its research. In contrast, the present study used a much larger and more diverse sample, which strengthens the reliability of the findings.

At the same time, some studies have reported no clear association between e-cigarette use and metabolic changes. This demonstrates that the evidence continues to develop. Overall, the current research study supports the theory that e-cigarette use increases the risk of metabolic disorders.

7 Real world application

According to the results, using e-cigarettes may be a risk factor for prediabetes. In actuality, during standard examinations, medical practitioners can examine e-cigarette users for early indicators of metabolic problems. Additionally, public health initiatives can increase knowledge of these hazards, particularly among young individuals. Furthermore, regulations targeted at lowering e-cigarette usage may aid in the prevention of type 2 diabetes (Zhang et al., 2022).

8 Future Research

A longitudinal study design should be used in future research. This will give a better understanding of whether e-cigarette use directly causes prediabetes rather than just being associated with it. Additionally, different aspects of uses like frequency, duration etc influence risk must also be studied. In addition, future studies should consider other factors like diet and lifestyle to provide a more complete understanding (Zhang et al., 2022).

9 Conclusion

The study shows that there exists a significant connection between adult e-cigarette consumption and increased prediabetes risk. The results show that e-cigarette use has major effects on metabolic health for people who do not smoke regular cigarettes. The study demonstrates that public education combined with early health assessments and preventive strategies serves as essential public health measures to decrease future health risks.

10 References

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